

ler: LVK (LIGO-Virgo-KAGRA collaboration) event (compact-binary mergers) rate calculator and simulator

Hemantakumar Phurailatpam ¹, Anupreeta More ^{2,3}, Harsh Narola ^{4,5}, Leo C. Y. Ng ^{1,6}, Justin Janquart ^{4,5,7,8}, Chris Van Den Broeck ^{4,5}, Otto Akseli Hannuksela ¹, Neha Singh ⁹, and David Keitel ⁹

¹ Department of Physics, The Chinese University of Hong Kong, Shatin, New Territories, Hong Kong ² The Inter-University Centre for Astronomy and Astrophysics (IUCAA), Post Bag 4, Ganeshkhind, Pune 411007, India ³ Kavli Institute for the Physics and Mathematics of the Universe (IPMU), 5-1-5 Kashiwanoha, Kashiwa-shi, Chiba 277-8583, Japan ⁴ Department of Physics, Utrecht University, Princetonplein 1, 3584 CC Utrecht, The Netherlands ⁵ Nikhef – National Institute for Subatomic Physics, Science Park, 1098 XG Amsterdam, The Netherlands ⁶ IGC, Institute for Gravitation and the Cosmos, Pennsylvania State University, University Park, PA 16802, USA ⁷ Center for Cosmology, Particle Physics and Phenomenology - CP3, Universit'e Catholique de Louvain, Louvain-La-Neuve, B-1348, Belgium ⁸ Royal Observatory of Belgium, Avenue Circulaire, 3, 1180 Uccle, Belgium ⁹ IAC3, Universitat de les Illes Balears, Crta. Valldemossa km 7.5, E-07122 Palma, Spain

DOI: [10.21105/joss.07420](https://doi.org/10.21105/joss.07420)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Warrick Ball](#)  

Reviewers:

- [@sibirrer](#)
- [@Jammy2211](#)

Submitted: 16 August 2024

Published: 03 August 2026

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Gravitational waves (GWs) are ripples in spacetime produced by accelerating massive objects. Since 2015, the LIGO-Virgo-KAGRA (LVK) collaboration ([Acernese et al., 2014](#); [Akutsu et al., 2020](#); [The LIGO Scientific Collaboration et al., 2015](#)) has routinely detected GW transients from compact-binary mergers, including binary black holes (BBHs), binary neutron stars (BNSs), and neutron star-black hole pairs (NSBHs), during observing runs ([B. P. Abbott et al., 2019](#); [R. Abbott et al., 2021, 2023](#); [The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, 2026](#)). Most of these signals reach detectors unobstructed, forming a baseline unlensed population, but a small fraction can encounter massive intervening objects such as galaxies or galaxy clusters. These objects act as lenses, splitting the signals into multiple images that are magnified, time-shifted, and phase-shifted through strong gravitational lensing ([Abbott et al., 2021](#); [Wierda et al., 2021](#)).

Accurate modeling of unlensed and lensed gravitational-wave populations is essential for astrophysical interpretation, inferring merger rate histories, and validating future lensing events ([Abbott et al., 2021](#); [Janquart et al., 2023](#); [The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, 2024](#); [Wierda et al., 2021](#)). The `ler` software is a Python framework that provides a unified pipeline to model these populations statistically and calculate their detectable rates. Producing precise results requires computationally demanding tasks such as large-scale sampling of source and lens attributes, solving lens equations, and simulating detector responses. `ler` reduces these costs through optimization and parallelization, achieving up to a thousandfold speedup relative to an un-optimized baseline implementation (further details provided in the Software design section). Source code, validation examples, and documentation are available through the project repository, [supporting examples](#), and [documentation site](#).

Statement of need

ler is a statistics-based Python tool designed for the joint simulation and computation of detectable rates for both unlensed and lensed gravitational-wave events. It is intended for LVK analyses and for astrophysics researchers working on compact-binary populations and strong lensing. Its core functionality integrates the sampling of compact-binary source properties and lens-galaxy characteristics, solving lens equations to derive image properties, and computing detectable event rates. This functionality leverages NumPy (Harris et al., 2020) for array operations and linear algebra alongside SciPy (Virtanen et al., 2020) interpolation methods and Python multiprocessing. Efficiency is improved using Numba (Lam et al., 2015) for just-in-time compilation and multithreading, and by using inverse-transform sampling and importance sampling where rejection sampling is inefficient. ler can use lenstronomy (Birrer et al., 2021) for lensing calculations and gwsnr (Phurailatpam & Hannuksela, 2025) for efficient detection-probability calculations P_{det} .

ler produces parameter distributions for intrinsic populations and for populations selected by detection criteria, including strong-lensing selection and image-level detectability. This supports validation studies for lensing candidates (Janquart et al., 2023) and forecasting for present and future detector networks (Ng et al., 2025). The design can be adapted to multi-messenger applications by changing the detection criteria, including studies of strongly lensed gamma-ray bursts (More & Phurailatpam, 2025). ler also supports integration with existing inference tools by providing P_{det} estimates that can be used to construct selection functions (see [supporting examples](#)) for hierarchical population inference frameworks (Thrane & Talbot, 2019) with Bilby software (Ashton et al., 2019).

State of the field

Joint population studies of unlensed and strongly lensed gravitational waves require end-to-end simulations that combine source and lens-galaxy models with detector selection effects. Common inference packages such as Bilby (Ashton et al., 2019) and cosmology and rate tools such as gwcsmo (Gray et al., 2020, 2022, 2023) provide general-purpose likelihood evaluation, sampling, and population inference utilities. However, they do not provide a single, configuration-driven pipeline to generate unlensed and strongly lensed populations jointly at scale, nor to propagate lensing image properties into detectability. Lens-modelling tools such as lenstronomy (Birrer et al., 2021) implement fast analytical routines for lensing calculations but are not designed as high-throughput population-level simulators.

Previous strong-lensing population studies often relied on pipelines tightly coupled to specific lens models or detection criteria (Haris et al., 2018; Wierda et al., 2021). Many studies report rate estimates based on limited sample sizes without showing convergence diagnostics or uncertainty estimates. They also commonly neglect uncertainty in population-model hyperparameters, which should propagate into the detectable population and rate predictions. ler addresses these gaps by providing an integrated, high-throughput workflow linking population draws, lensing cross-sections, image properties, and detector selection in one reproducible pipeline.

Software design

ler is designed as an end-to-end population simulator with a clear separation between the scientific steps and the computational machinery. The pipeline is organized around two rate calculations. Unlensed rates are computed by drawing compact-binary populations from astrophysical priors, evaluating detectability, and integrating over the detected events. Lensed rates extend this flow by sampling lens-galaxy and source-redshift parameters under a strong-lensing condition, solving the lens equation to generate image properties, applying detectability

criteria at the image level, and integrating over the detectable lensed events. This separation keeps the rate logic explicit while allowing the source, lens, and detection models to be replaced without rewriting the full pipeline. The mathematical foundations for these implementations are detailed in the Mathematics section.

A core design choice is modularity with stable interfaces between modules that handle source populations, lens populations, image-property calculations, and rate estimation. Users can select built-in models such as BBH, BNS, or NSBH populations and lens models such as singular isothermal sphere (SIS), singular isothermal ellipsoid (SIE), or elliptical power-law with external shear (EPL+Shear), or provide their own distributions and detection criteria. Detector selection is handled through a backend interface so that signal-to-noise and detection probability calculations can be delegated to `gwsnr` or replaced by alternative implementations. This architecture supports extension and testing while keeping the scientific assumptions visible in configuration rather than embedded in ad hoc scripts.

Performance is obtained by compiling the repeatedly evaluated parts of the pipeline with Numba `njit` and parallelizing the main Monte Carlo loops. `ler` uses `lenstronomy` (Birrer et al., 2021) as an optional backend, but it provides an in-house analytical EPL+Shear solver and caustic and cross-section calculator that are compiled with Numba and run in parallel, following the same mathematical formulation as `lenstronomy` and Tessore (2015) (Tessore & Benton Metcalf, 2015). This leads to two practical trade-offs. First, the rate estimates are Monte Carlo approximations, so accuracy is controlled by sample size rather than closed-form expressions, and `ler` reports convergence statistics from repeated batches. Second, just-in-time compilation adds overhead on first use, which is about 2 s for the first unlensed batch and about 25 s for the first lensed batch that includes image-property calculations. After compilation, the compiled kernels reduce per-sample cost and make large simulations feasible. In benchmarks that disable just-in-time compilation and parallelism, these choices provide speed gains of about $180\times$ for unlensed populations and $5189\times$ for lensed populations. On a standard six-core processor (Apple M2 Pro, 16GB RAM), a batch of 10^5 samples takes about 300 ms for unlensed calculations and about 30 s for lensed calculations, and repeated batches converge to a stable rate estimate.

Mathematics

The mathematical workflow in `ler` can be summarised as Monte Carlo estimates of detectable event rates for unlensed and strongly lensed populations, obtained by averaging detection probabilities over simulated events drawn from the relevant population models. For unlensed events, the detector-frame rate of detectable events is

$$\frac{\Delta N_{\text{U}}^{\text{obs}}}{\Delta t} = \mathcal{N}_{\text{U}} \left\langle P(\text{obs} \mid \vec{\theta}) \right\rangle_{\vec{\theta} \sim P(\vec{\theta})}$$

where $\mathcal{N}_{\text{U}}$ is the total intrinsic merger rate in the detector frame, $\vec{\theta}$ denotes the source parameters drawn from $P(\vec{\theta})$, and $P(\text{obs} \mid \vec{\theta})$ is the detection probability.

For strongly lensed events, the detector-frame rate of detectable lensed events is

$$\frac{\Delta N_{\text{L}}^{\text{obs}}}{\Delta t} = \mathcal{N}_{\text{L}} \left\langle P(\text{obs} \mid \vec{\theta}_{\text{U}}, \vec{\theta}_{\text{L}}, \vec{\beta}, \text{SL}) \right\rangle_{\substack{\vec{\theta}_{\text{U}}, \vec{\theta}_{\text{L}} \sim P(\vec{\theta}_{\text{U}}, \vec{\theta}_{\text{L}} \mid z_{\text{L}}, z_{\text{s}}, \text{SL}) \\ \vec{\beta} \sim P(\vec{\beta} \mid z_{\text{s}}, \vec{\theta}_{\text{L}}, \text{SL})}},$$

where $\mathcal{N}_{\text{L}}$ is the total intrinsic merger rate in the detector frame for the lensed population, $\vec{\theta}_{\text{U}}$ and $\vec{\theta}_{\text{L}}$ denote the source and lens parameters, $\vec{\beta}$ is the source position in the source plane, and z_{L} and z_{s} are the lens and source redshifts, respectively. The distributions

$P(\vec{\theta}_U, \vec{\theta}_L | z_L, z_s, \text{SL})$ and $P(\vec{\beta} | z_s, \vec{\theta}_L, \text{SL})$ define the hierarchical sampling under the strong-lensing condition SL, and $P(\text{obs} | \vec{\theta}_U, \vec{\theta}_L, \vec{\beta}, \text{SL})$ is evaluated using the image properties implied by $(\vec{\theta}_L, \vec{\beta})$ and the chosen detectability criteria.

Research impact statement

ler has established a measurable impact within the gravitational-wave community through its integration into research workflows and its growing user base. The package supports a diverse community ranging from students to faculty members at multiple institutions, with development driven by researchers across various organizations and [peer-reviewed within the LVK Scientific Collaboration](#). Active community engagement is maintained through bug reports and feature requests on the GitHub issue tracker and through collaboration communication channels such as LIGO Mattermost.

The software has facilitated several scientific publications related to gravitational-wave lensing statistics. It has been used to analyze the distribution of time delays and magnification ratios for the validation of lensing candidates ([Janquart et al., 2023](#)) and to forecast the detection of sub-threshold lensed events with future observatories ([Ng et al., 2025](#)). Additional research applications include the generation of lensed-event statistics and the calculation of posterior odds ([Hannuksela et al., 2026](#)), as well as forecasting electromagnetic counterparts for strongly lensed gamma-ray bursts ([More & Phurailatpam, 2025](#)). These implementations demonstrate that the tool provides a robust framework for high-level population studies and astrophysical forecasting.

AI usage disclosure

No generative AI tools were used to develop the software, write this manuscript, or prepare the accompanying materials.

Acknowledgements

The authors thank their academic advisors for guidance and support. Hemantakumar Phurailatpam acknowledges the Department of Physics at The Chinese University of Hong Kong for the Postgraduate Studentship that facilitated this research. Hemantakumar Phurailatpam and Otto A. Hannuksela acknowledge support from the Research Grants Council of Hong Kong (Project Nos. CUHK 14304622 and 14307923), the start-up grant from The Chinese University of Hong Kong, and the Direct Grant for Research from the Research Committee of The Chinese University of Hong Kong. The authors also acknowledge support from the Netherlands Organisation for Scientific Research (NWO). N. Singh is supported through the Conselleria d'Educació i Universitats del Govern de les Illes Balears via a Vicenç Mut postdoctoral grant (POSTDOC2024_55) with funds from the European Social Fund+ in the framework of the Balearic Islands ESF+ Program 2021-2027. N. Singh and D. Keitel are supported by the Universitat de les Illes Balears (UIB) with funds from the Programa de Foment de la Recerca i la Innovació de la UIB 2024-2026 (supported by the yearly plan of the Tourist Stay Tax ITS2023-086); the Spanish Agencia Estatal de Investigación grants PID2022-138626NB-I00, RED2024-153978-E, RED2024-153735-E, funded by MICIU/AEI/10.13039/501100011033 and the ERDF/EU; and the Comunitat Autònoma de les Illes Balears through the Conselleria d'Educació i Universitats with funds from the European Union - European Regional Development Fund (ERDF) (SINCO2022/18146 - Plataforma HiTech-IAC3-BIO). The authors acknowledge the computational resources provided by the LIGO Laboratory, supported by National Science Foundation Grants No. PHY-0757058 and No. PHY-0823459.

References

- Abbott, R., Abbott, T. D., Abraham, S., Acernese, F., Ackley, K., Adams, A., Adams, C., Adhikari, R. X., Adya, V. B., Affeldt, C., Agarwal, D., Agathos, M., Agatsuma, K., Aggarwal, N., Aguiar, O. D., Aiello, L., Ain, A., Ajith, P., Aleman, K. M., ... Zweizig, J. (2021). Search for lensing signatures in the gravitational-wave observations from the first half of LIGO–virgo’s third observing run. *The Astrophysical Journal*, 923(1), 14. <https://doi.org/10.3847/1538-4357/ac23db>
- Acernese, F., Agathos, M., Agatsuma, K., Aisa, D., Allemandou, N., Allocca, A., Amarni, J., Astone, P., Balestri, G., Ballardin, G., Barone, F., Baronick, J.-P., Barsuglia, M., Basti, A., Basti, F., Bauer, T. S., Bavigadda, V., Bejger, M., Beker, M. G., ... Zendri, J.-P. (2014). Advanced virgo: A second-generation interferometric gravitational wave detector. *Classical and Quantum Gravity*, 32(2), 024001. <https://doi.org/10.1088/0264-9381/32/2/024001>
- Akutsu, T., Ando, M., Arai, K., Arai, Y., Araki, S., Araya, A., Aritomi, N., Aso, Y., Bae, S., Bae, Y., Baiotti, L., Bajpai, R., Barton, M. A., Cannon, K., Capocasa, E., Chan, M., Chen, C., Chen, K., Chen, Y., ... Zhu, Z.-H. (2020). Overview of KAGRA: Detector design and construction history. *Progress of Theoretical and Experimental Physics*, 2021(5), 05A101. <https://doi.org/10.1093/ptep/ptaa125>
- Ashton, G., Hübner, M., Lasky, P. D., Talbot, C., Ackley, K., Biscoveanu, S., Chu, Q., Divakarla, A., Easter, P. J., Goncharov, B., Vivanco, F. H., Harms, J., Lower, M. E., Meadors, G. D., Melchor, D., Payne, E., Pitkin, M. D., Powell, J., Sarin, N., ... Thrane, E. (2019). Bilby: A user-friendly bayesian inference library for gravitational-wave astronomy. *The Astrophysical Journal Supplement Series*, 241(2), 27. <https://doi.org/10.3847/1538-4365/ab06fc>
- B. P. Abbott et al. (2019). GWTC-1: A gravitational-wave transient catalog of compact binary mergers observed by LIGO and virgo during the first and second observing runs. *Physical Review X*, 9(3). <https://doi.org/10.1103/physrevx.9.031040>
- Birrer, S., Shajib, A. J., Gilman, D., Galan, A., Aalbers, J., Millon, M., Morgan, R., Pagano, G., Park, J. W., Teodori, L., Tessore, N., Ueland, M., Vyvere, L. V. de, Wagner-Carena, S., Wempe, E., Yang, L., Ding, X., Schmidt, T., Sluse, D., ... Amara, A. (2021). Lenstronomy II: A gravitational lensing software ecosystem. *Journal of Open Source Software*, 6(62), 3283. <https://doi.org/10.21105/joss.03283>
- Gray, R., Beirnaert, F., Karathanasis, C., Revenu, B., Turski, C., Chen, A., Baker, T., Vallejo, S., Enea Romano, A., Ghosh, T., Ghosh, A., Leyde, K., Mastrogiovanni, S., & More, S. (2023). Joint cosmological and gravitational-wave population inference using dark sirens and galaxy catalogues. *Journal of Cosmology and Astroparticle Physics*, 2023(12), 023. <https://doi.org/10.1088/1475-7516/2023/12/023>
- Gray, R., Magaña Hernandez, I., Qi, H., Sur, A., Brady, P. R., Chen, H.-Y., Farr, W. M., Fishbach, M., Gair, J. R., Ghosh, A., Holz, D. E., Mastrogiovanni, S., Messenger, C., Steer, D. A., & Veitch, J. (2020). Cosmological inference using gravitational wave standard sirens: A mock data analysis. *Physical Review D*, 101(12), 122001. <https://doi.org/10.1103/PhysRevD.101.122001>
- Gray, R., Messenger, C., & Veitch, J. (2022). A pixelated approach to galaxy catalogue incompleteness: Improving the dark siren measurement of the hubble constant. *Monthly Notices of the Royal Astronomical Society*, 512(1), 1127–1140. <https://doi.org/10.1093/mnras/stac366>
- Hannuksela, O. A., Haris, K., Janquart, J., Narola, H., Phurailatpam, H., Creighton, J. D. E., & Van Den Broeck, C. (2026). Strong gravitational-wave lensing posterior odds. *The Astrophysical Journal*, 1002(1), 42. <https://doi.org/10.3847/1538-4357/ae5816>
- Haris, K., Mehta, A. K., Kumar, S., Venumadhav, T., & Ajith, P. (2018). *Identifying strongly lensed gravitational wave signals from binary black hole mergers*. <https://doi.org/10.48550/>

[arXiv.1807.07062](https://arxiv.org/abs/1807.07062)

- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., Fernández del Río, J., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Janquart, J., Wright, M., Goyal, S., Chan, J. C. L., Ganguly, A., Garrón, Á., Keitel, D., Li, A. K. Y., Liu, A., Lo, R. K. L., Mishra, A., More, A., Phurailatpam, H., Prasia, P., Ajith, P., Biscoveanu, S., Cremonese, P., Cudell, J. R., Ezquiaga, J. M., ... Veitch, J. (2023). Follow-up analyses to the O3 LIGO–Virgo–KAGRA lensing searches. *Monthly Notices of the Royal Astronomical Society*, 526(3), 3832–3860. <https://doi.org/10.1093/mnras/stad2909>
- Lam, S. K., Pitrou, A., & Seibert, S. (2015). Numba: A LLVM-based python JIT compiler. *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, 1–6. <https://doi.org/10.1145/2833157.2833162>
- More, A., & Phurailatpam, H. (2025). Gravitational lensing: Towards combining the multi-messengers. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2295), 20240127. <https://doi.org/10.1098/rsta.2024.0127>
- Ng, L. C. Y., Janquart, J., Phurailatpam, H., Narola, H., Poon, J. S. C., Van Den Broeck, C., & Hannuksela, O. A. (2025). Uncovering faint lensed gravitational-wave signals and reprioritizing their follow-up analysis using galaxy lensing forecasts with detected counterparts. *Monthly Notices of the Royal Astronomical Society*, 540(4), 2937–2951. <https://doi.org/10.1093/mnras/staf874>
- Phurailatpam, H., & Hannuksela, O. A. (2025). *Gwsnr: A python package for efficient signal-to-noise calculation of gravitational-waves*. <https://doi.org/10.48550/arXiv.2412.09888>
- R. Abbott et al. (2021). GWTC-2: Compact binary coalescences observed by LIGO and virgo during the first half of the third observing run. *Physical Review X*, 11(2), 021053. <https://doi.org/10.1103/PhysRevX.11.021053>
- R. Abbott et al. (2023). GWTC-3: Compact binary coalescences observed by LIGO and virgo during the second part of the third observing run. *Physical Review X*, 13(4), 041039. <https://doi.org/10.1103/PhysRevX.13.041039>
- Tessore, N., & Benton Metcalf, R. (2015). The elliptical power law profile lens. *Astronomy and Astrophysics*, 580, A79. <https://doi.org/10.1051/0004-6361/201526773>
- The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration. (2024). Search for gravitational-lensing signatures in the full third observing run of the LIGO–Virgo network. *The Astrophysical Journal*, 970(2), 191. <https://doi.org/10.3847/1538-4357/ad3e83>
- The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration. (2026). GWTC-4.0: Updating the gravitational-wave transient catalog with observations from the first part of the fourth LIGO–Virgo–KAGRA observing run. *The Astrophysical Journal Letters*, 1004(2), L22. <https://doi.org/10.3847/2041-8213/ae2c74>
- The LIGO Scientific Collaboration, Aasi, J., Abbott, B. P., Abbott, R., Abbott, T., Abernathy, M. R., Ackley, K., Adams, C., Adams, T., Addesso, P., Adhikari, R. X., Adya, V., Affeldt, C., Aggarwal, N., Aguiar, O. D., Ain, A., Ajith, P., Alesic, A., Allen, B., ... Zweizig, J. (2015). Advanced LIGO. *Classical and Quantum Gravity*, 32(7), 074001. <https://doi.org/10.1088/0264-9381/32/7/074001>
- Thrane, E., & Talbot, C. (2019). An introduction to bayesian inference in gravitational-wave astronomy: Parameter estimation, model selection, and hierarchical models. *Publications of the Astronomical Society of Australia*, 36. <https://doi.org/10.1017/pasa.2019.2>

- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17(3), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Wierda, A. R. A. C., Wempe, E., Hannuksela, O. A., Koopmans, L. V. E., Agnello, A., Bonvin, C., Bucciarelli, B., Camera, C., Czoske, O., Finke, C., & others. (2021). Beyond the detector horizon: Forecasting gravitational-wave strong lensing. *The Astrophysical Journal*, 921(1), 154. <https://doi.org/10.3847/1538-4357/ac1bb4>